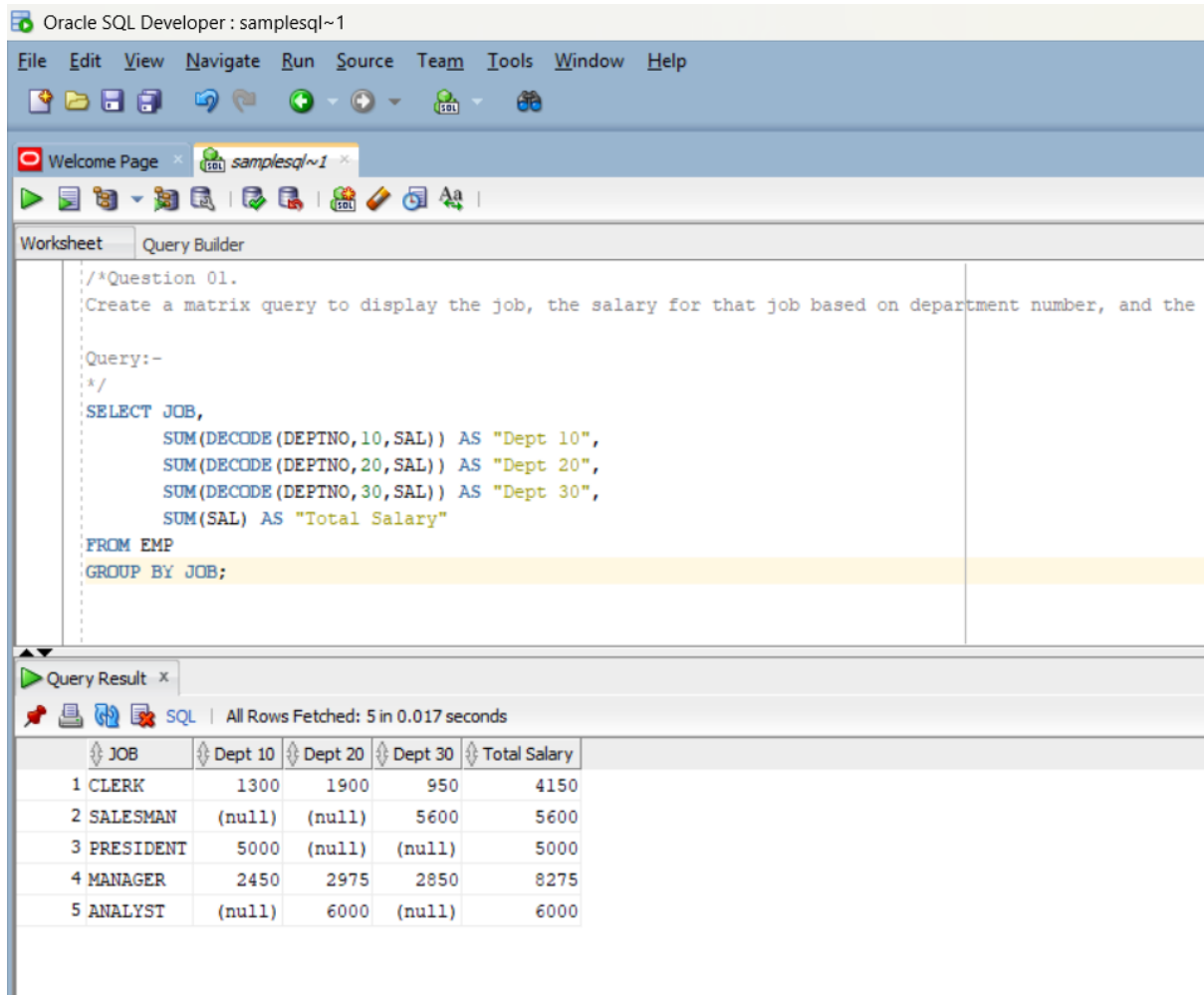


# Using Set Operators

Handson Assessment Questions for using set operators---

01.



The screenshot displays the Oracle SQL Developer interface. The main window shows a SQL script in the 'Query Builder' tab. The script is a matrix query designed to display job roles, salaries for specific departments (10, 20, 30), and a total salary for each job. The query uses the DECODE function to pivot department data into columns.

```
/*Question 01.  
Create a matrix query to display the job, the salary for that job based on department number, and the  
  
Query:-  
*/  
SELECT JOB,  
       SUM(DECODE(DEPTNO,10,SAL)) AS "Dept 10",  
       SUM(DECODE(DEPTNO,20,SAL)) AS "Dept 20",  
       SUM(DECODE(DEPTNO,30,SAL)) AS "Dept 30",  
       SUM(SAL) AS "Total Salary"  
FROM EMP  
GROUP BY JOB;
```

Below the query editor, the 'Query Result' window shows the output of the executed query. It indicates that all 5 rows were fetched in 0.017 seconds. The results are presented in a table with 5 columns: JOB, Dept 10, Dept 20, Dept 30, and Total Salary.

| JOB         | Dept 10 | Dept 20 | Dept 30 | Total Salary |
|-------------|---------|---------|---------|--------------|
| 1 CLERK     | 1300    | 1900    | 950     | 4150         |
| 2 SALESMAN  | (null)  | (null)  | 5600    | 5600         |
| 3 PRESIDENT | 5000    | (null)  | (null)  | 5000         |
| 4 MANAGER   | 2450    | 2975    | 2850    | 8275         |
| 5 ANALYST   | (null)  | 6000    | (null)  | 6000         |

02.

Worksheet Query Builder

QUESTION 02.

Using set operator display the DEPTNO,SUM(SAL) for each dept, JOB,SUM(SAL) for each Job and Total Salary.

Query:-

```
*/
SELECT TO_CHAR(DEPTNO) AS CATEGORY,
       SUM(SAL) AS TOTAL_SALARY
FROM EMP
GROUP BY DEPTNO

UNION ALL

SELECT JOB,
       SUM(SAL)
FROM EMP
GROUP BY JOB

UNION ALL

SELECT 'TOTAL',
       SUM(SAL)
FROM EMP;
```

Query Result x

SQL | All Rows Fetched: 9 in 0.003 seconds

| CATEGORY    | TOTAL_SALARY |
|-------------|--------------|
| 1 30        | 9400         |
| 2 20        | 10875        |
| 3 10        | 8750         |
| 4 CLERK     | 4150         |
| 5 SALESMAN  | 5600         |
| 6 PRESIDENT | 5000         |
| 7 MANAGER   | 8275         |
| 8 ANALYST   | 6000         |
| 9 TOTAL     | 29025        |

03.

Using Set Operator display the JOB and Deptno in employees working in deptno 20,10,30 in that order.

Query:-

```
*/
SELECT JOB, DEPTNO
FROM EMP
WHERE DEPTNO = 20

UNION ALL

SELECT JOB, DEPTNO
FROM EMP
WHERE DEPTNO = 10

UNION ALL

SELECT JOB, DEPTNO
FROM EMP
WHERE DEPTNO = 30;
```

Query Result x

SQL | All Rows Fetched: 14 in 0.006 seconds

| JOB         | DEPTNO |
|-------------|--------|
| 1 CLERK     | 20     |
| 2 MANAGER   | 20     |
| 3 ANALYST   | 20     |
| 4 CLERK     | 20     |
| 5 ANALYST   | 20     |
| 6 MANAGER   | 10     |
| 7 PRESIDENT | 10     |
| 8 CLERK     | 10     |
| 9 SALESMAN  | 30     |
| 10 SALESMAN | 30     |